

Institute of Actuaries of India

Subject CT2 – Finance and Financial Reporting

March 2018 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

<u>Solution 1:</u> D	[2 Marks]
<u>Solution 2:</u> B	[2 Marks]
<u>Solution 3:</u> D	[2 Marks]
<u>Solution 4:</u> C	[2 Marks]
<u>Solution 5:</u> B	[2 Marks]
<u>Solution 6:</u> A	[2 marks]
<u>Solution 7:</u> D	[2 marks]
<u>Solution 8:</u> B	[2 Marks]
<u>Solution 9:</u> C	[2 Marks]
<u>Solution 10:</u> C	[2 Marks]

Solution 11:

Modigliani and Miller's first irrelevance proposition: The market value of any firm is independent of its capital structure. Modigliani and Miller's second irrelevance proposition: The expected rate of return on the company's shares increases in proportion to the debt-equity ratio, expressed in market values.

[2 marks]

Solution 12:

i)

A listing allows the company to sell new shares to a wide market and so to **raise capital** at the point of listing (as the company is likely to choose a method of obtaining a listing that raises capital), and in the future from an expanded policyholder base.

Listing will also make it easier to **raise short- and long-term debt finance in future**, because lenders feel safer and happier about lending money to a listed company (since it has to meet the exchange's initial and on-going requirements - Compliance).

Listing on a stock exchange will give existing shareholders a **convenient exit route**, e.g. venture capitalists who supported the company in its early years and wish to realise their investment.

A listing makes the **company's shares more marketable** and more accurately and easily valued.

A listing will also make **employee share participation and/or director share option schemes more attractive**.

Listing may **improve the status of the company and increase public awareness of the Company / industry** leading to increased sales.

In the **Indian context**, companies are getting high valuations due to significant untapped potential of the insurance market.

(5)

ii) **Costs** associated with listing

Dilution of control

Unwelcome takeover bid hard to avoid

Disadvantages **more relevant to family owned businesses**

(2)

iii) SBI Life / HDFC Life / ICICI Prudential Life

Offer for sale / Initial Public Offering (IPO)

(2)

[9 marks]

Solution 13:

i) A panic sell-off by small retail investors is likely to reduce share prices.

(1)

ii) Positive short-term sentiment is likely to increase share price.

(1)

iii) If the business was already quite highly geared before the additional borrowing, shareholders may become concerned about whether or not the company can continue to service the interest burden. In order to carry that risk, investors may require a higher prospective return – i.e. they will buy the shares at a lower price. The business is relatively stable with low growth and low profit volatility. Investors may be happy to have the additional gearing. It gives them a higher prospective return immediately (the project adds value) and gives them increased exposure to any upside risks for the company. In the event of a wind-up, the assets of the company would first be used to repay the bondholders. If they are concerned about the reduced capital cover, the equity price may fall.

(2)

[4 marks]

Solution 14:

Five shares held before the issue will be worth $5 \times \text{Rs}218 = \text{Rs}1090$.

The new project will increase market capitalisation by 20%, so five shares will increase in value to $\text{Rs}1090 + 20\% = \text{Rs}1308$.

The cash injection will add a further Rs185 per five shares = $\text{Rs}1308 + \text{Rs}185 = \text{Rs}1493$.

Each share will be worth $\text{Rs}1493/6 = \text{Rs}248.83$.

Yes the rights issue appears to be beneficial as the market price of the share is higher post rights issue.

[5 marks]

Solution 15:

i)

We can compare individual stocks with the overall market by assessing the *beta* (β) of the stock.

For company *i* :

$$\beta_i = \frac{\sigma_{im}}{\sigma_m^2}$$

where σ_m^2 is the variance of the market index and σ_{im} the covariance between the individual stock's return and that of the market.

(1)

ii) Different industry – inherent riskiness

Geared Beta might be different because of different debt ratio Brand value
Some stocks are used more for speculative purposes hence more volatile.

(1)

iii) Geared beta = Ungeared beta + (1+D/E * (1-Tax))

$$1.1 = \text{Ungeared beta} + (1 + 1/2 * .7)$$

$$\text{Ungeared beta} = 0.8148$$

$$\text{New geared beta} = 0.8148 + (1 + 2/2 * .7) = 1.385$$

(3)

[5 marks]**Solution 16:**

i)

Year	Cashflows			Discount factor	Discounted cashflows		
	Option 1	Option 2	Option 3		(20,919)	60,964	13,287
1	-1000000	-200000	- 1000000	1	(1,000,000)	(200,000)	(1,000,000)
2		-200000		0.93457943	-	(186,916)	-
3		-200000		0.87343872	-	(174,688)	-
4		-200000	500000	0.81629787	-	(163,260)	408,149
5		-200000		0.76289521	-	(152,579)	-
6		300000		0.71298617	-	213,896	-
7		300000	500000	0.66634222	-	199,903	333,171
8		300000		0.62274974	-	186,825	-

9		300000		0.58200910	-	174,603	-
10	1800000	300000	500000	0.54393374	979,081	163,180	271,967
Total	800,000	500,000	500,000		(20,919)	60,964	13,287

Investment option 2 is the best option using an NPV approach, with a view to prevailing risk-free rates in the market. (5)

ii) a)

Year	Option 4 CF	Discount factor	Option 4 discounted CF
1	-200000	1	(200,000)
2	-200000	0.925925926	(185,185)
3	-200000	0.85733882	(171,468)
4	-200000	0.793832241	(158,766)
5	-200000	0.735029853	(147,006)
6	270000	0.680583197	183,757
7	291600	0.630169627	183,757
8	314928	0.583490395	183,757
9	340,122	0.540268885	183,757
10	367,332	0.500248967	183,757
Total	583,982		56,362

Option 2 still is the better option using an NPV approach with appropriate RDR. (3)

ii) b) The other approach to assess risk in Option 4 is to use a sensitivity testing approach. Various scenarios could be tested looking at the past performance of the relevant index over different time periods. A weighted average NPV can be derived using the probability of occurrence, rather than making a straight adjustment in the RDR.

(2)

iii) Lock-in / payback period – How soon the investment returns initial capital and the period of time the money is unavailable for use.

Liquidity needs – Whether income is needed at certain time points

Tax considerations – Tax benefits under different options. E.g. under Option 2, money invested second year onwards may be taxed if not invested immediately

Internal Rate of Return

(3)

[13 marks]

Solution 17:

i) Managing interest rate risk / guaranteed liability profile

(1)

ii) Cost benefit tradeoff, interest rate outlook, risks involved (mainly counterparty risk)

(1)

[2 marks]

Solution 18:

i) The reason of complication in preparing the insurance companies accounts as compared to normal company accounts are as follows –

a) The insurance contracts with policyholders fall due outside the accounting period and are uncertain in size.

b) Premature transfer of profit to shareholders may endanger the financial stability of the company and ability to meet the future liabilities.

(1)

ii) The major component of reserves for a general insurance company's technical accounts are as follows-

a) URR (Un-expired risk reserves)

b) Outstanding claim reserves

c) URR (Un-expired risk reserves): This reserves is kept to cover the claims and expense that are expected to emerge from an unexpired portion of the insurance cover in future period

d) Outstanding claim reserves: This reserves is kept to cover the claims and expense for all outstanding claims that have not yet been settled. These claims and expense may or may not be reported or known at the time of preparation of accounts.

(4)

[5 marks]

Solution 19:

i) Company ABC acquires 100% share of Company XYZ. XYZ company share capital is worth of 2,00,000 and the par value of share is INR20.

So, the number of XYZ shares are = Share Capital/ Par value of XYZ share =
 $2,00,000/20=10,000$

ABC has to offer 5,000 share (1 share in ABC for every 2 shares in XYZ) which cost ABC =
 $5,000 \times 80 = \text{INR } 4,00,000$
 Total value of XYZ to ABC will be sum of XYZ's Share capital and reserves
 i.e $2,00,000 + 50,000 = 2,50,000$
 The good will cost of control for ABC will be = $4,00,000 - 2,50,000 = \text{INR } 1,50,000$.

(2)

ii) If Company ABC acquires 50% share of Company XYZ.

So, ABC is planning to acquire 5,000 shares of XYZ (Total shares of XYZ*50%) ABC has to offer 2,500 share which cost ABC = $2,500 \times 80 = \text{INR } 2,00,000$
 Total value of XYZ to ABC will be 50% of sum of XYZ's Share capital and reserves i.e
 $(2,00,000 + 50,000) \times 50\% = 1,25,000$
 The good will cost of control for ABC will be = $2,00,000 - 1,25,000 = \text{INR } 75,000$

(2)

[4 marks]

Solution 20:

i) According to realization concepts, Income and profits should be recognized as and when it is earned. According to accrual concepts, expenses should be recognized as and when they are incurred regardless of whether or not the amount has been paid.

(1)

ii) The earned premium for the year is calculated by considering the earned portion of the total written premium in the year and carried forward unexpired risk premium is an application of realization concept. The incurred claims for a financial year considers the paid claim during the year, outstanding claim reserves for known and un-known claims at end and start of the financial year.

(3)

[4 marks]

Solution 21:

Depreciation is defined as the measure of wearing out, consumption or other reduction in the useful economic life of a fixed asset, whether arising from passage of time and/or obsolescence through technological or market change. Depreciation adjustment can be done by two methods.

The Straight line basis: This method charges equal amount every years as $(\text{Cost} - \text{Estimated residual Value}) / \text{Estimated useful life}$.

The reducing balance method: This method charges a fixed percentage of "book value" (cost less depreciation to date) each year as $[1 - (\text{Estimated residual value} / \text{Cost})^{(1/\text{estimated useful life in years})}]$.

[3 marks]

Solution 22:

1. Ratio analysis does not consider the size of the company and can divert the attention from figures and statements.
2. Ratio analysis may not be useful in presenting appropriate comparison due to involvement of different accounting practice and external factors.
3. Ratio analysis may not show the true picture if there is an opportunity for management to apply bias towards few accounting policies and assumption which is called creative accounting.
4. The peculiarities of trade may make ratio analysis less useful because of difficulty to interpret few certain ratio.

[4 marks]**Solution 23:**

Moon Healthcare accounts will be as follows:

Expense Revenue account for the year ended 31st March 2017

Expense

Particulars	Amount (in INR million)	
Claims under policies		
Less recovered from reinsurance	580	
Add: out at the end of the year	100	680
Commission on direct business		300
Commission to broker for placing reinsurance		10
Expense of Management		300
Profit transferred to P&L account		1,710
Reserves for unexpired risk 50% of net premium		1,950
		4,950

Revenue

Particulars	Amount (in INR millions)
Balance of account at the beginning of the year	
Reserve for unexpired reserves (Health Fund @ start of FY 2017)	1,000
Premium less reinsurance	3,900
Commission on re-insurance ceded	50
	4,950**

Profit and Loss account for the year ended 31st March 2017

Particulars	Amount (in INR millions)
Rents	60
Audit fees	40
Provision for taxes	528
Transfer to general reserve	200
Balance c/d (carried forward)	1,082
	1,910

Particulars	Amount (in INR millions)
Balance b/d (brought forward)	50
Profit transferred from revenue account	1710
Income from investments	150
	1,910**

Balance sheet as on 31st March 2017:

Liabilities	Amount (in INR million)	
Share capital		900
General reserves	50	
Additions during the year	200	250
Profit and loss account		1,082
Health revenue account		1,950
Other liabilities		
Outstanding claim reserves including IBNR	100	
Provision for taxation	528	
Sundry creditor	25	653
		4,835

Assets	Amount (in INR million)
Investments	4,615
Outstanding premium	20
Cash and bank balances	200
	4,835**

Notes:

1)	Claims paid	600	
	Less: Recover from reinsurer	20	580
2)	Premium from direct business	4,000	
	Less: Reinsurance premium	100	3,900
4)	Reserves for unexpired risk, 50% of net premium		1,950
5)	Provision for taxation		
	Add: Profit from revenue account	1,710	
	Add: Income from investments	150	
		1,860	
	Less: Rents	60	
	Less: Audit fees	40	
		100	1,760
	Provision for taxes@30%		528

[20 marks]
